Alagille Syndrome, Part Eight: Ileal Bile Acid Transporter Inhibitors.

Maralixibat in the ICONIC Trial, Odevixibat in the ASSERT Trial, Event-Free Survival, and When to Start.

Imagine a girl of two with Alagille syndrome, confirmed by a disease-causing variant in the gene JAG1. She still has her own liver. Her total bilirubin is about four milligrams per decilitre, and her serum bile acids are around two hundred and ten micromoles per litre, many times the upper limit of normal. She takes ursodeoxycholic acid, a bile acid given to make bile flow more freely, and rifampicin, the best studied of the older drugs for cholestatic itch. Even so, her mother's itch diary averages a little under three on a scale from zero to four. Small yellow plaques of cholesterol, called xanthomas, are beginning to appear over her knuckles, and her height sits almost two standard deviations below the average for her age.

The team has five questions. There are now two approved drugs of a new class, the ileal bile acid transporter inhibitors, usually shortened to IBAT inhibitors. Is there any reason to prefer one, maralixibat or odevixibat, and at what dose? Should she start now, or only after every older drug has failed and the family has stopped sleeping? Will the drug change the long-term fate of her liver, or only her skin? What needs to be checked for safety, and when should the drug be stopped? And if the itch has not moved after three months, what then?

The main idea of this chapter is that IBAT inhibitors are the first drugs with randomised evidence for itch in Alagille syndrome. For maralixibat the evidence is a randomised withdrawal trial, and for odevixibat it is a conventional placebo-controlled trial. Both drugs lower serum bile acids, and both lower the itch score by roughly one to two points on a zero to four scale. Data reaching six and seven years suggest that children treated with maralixibat have fewer serious liver events and fewer transplants, but that conclusion comes from a comparison with a matched historical cohort, not from a randomised trial. The 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver diseases says that an IBAT inhibitor should be offered, when available, to patients with Alagille syndrome and cholestatic itch. In practice that means starting the drug in the itchy child whose liver is still compensated, rather than saving it for the end of the ladder, and watching liver tests and fat-soluble vitamins on purpose.

How blocking a gut transporter empties the bile acid pool

To understand these drugs, start with the enterohepatic circulation. The liver makes bile acids from cholesterol and secretes them into bile. Bile acids travel down the bile ducts into the small intestine, where they help absorb fat. They are not lost in the stool. About ninety-five percent of them are taken back up at the end of the small intestine, the terminal ileum, by a protein called the apical sodium-dependent bile acid transporter. This is the same protein as the ileal bile acid transporter, and its gene is called SLC10A2. From the ileum the bile acids return through the portal vein to the liver, which takes them up and secretes them again. The whole pool cycles several times every day.

In a cholestatic child, bile cannot leave the liver normally, so bile acids build up in the liver cells and spill into the blood and the skin. Part of that load is bile acid that has been reabsorbed from the gut and recycled. Maralixibat and odevixibat bind the transporter on the inside surface of the ileum. They are minimally absorbed into the body, so they act almost entirely in the gut lumen. With the transporter blocked, more bile acid passes on into the colon and leaves in the stool, and the pool that returns to the liver shrinks.

There is a feedback loop worth understanding step by step. When bile acids enter the ileal cells, those cells release a hormone called fibroblast growth factor nineteen, which travels to the liver and tells it to slow down the making of new bile acids. When the transporter is blocked, less bile acid enters the ileal cells, so less of this hormone is released. The liver responds by making more bile acid from cholesterol, and a blood marker of that synthesis, called C4, rises. Even so, the net effect is a fall in serum bile acids, because the losses in the stool outweigh the extra synthesis. Because the extra bile acid is made from cholesterol, many children also see their cholesterol fall and their xanthomas slowly regress.

Two practical consequences follow from this mechanism. The first is that the drug needs bile in the gut. If no bile reaches the intestine, as after a failed Kasai operation for biliary atresia or in complete obstruction, there is nothing for the drug to act on. In Alagille syndrome bile flow is reduced but usually present, which is why the drug can work. The second is that this is, in effect, a medical version of surgical biliary diversion. It targets the same mediators that cholestyramine tries to bind, but upstream and more completely. The price is more bile acid in the colon, which is why diarrhoea and loss of fat-soluble vitamins are the effects of the whole class.

Two drugs, two trials, two labels

Two IBAT inhibitors are approved for itch in Alagille syndrome. Maralixibat is sold as Livmarli. In the United States it is indicated for cholestatic itch in Alagille syndrome from three months of age. The labelled dose starts at one hundred and ninety micrograms per kilogram once daily for one week and then rises to three hundred and eighty micrograms per kilogram once daily, taken thirty minutes before breakfast. It comes as an oral solution, and as tablets for children who weigh twenty-five kilograms or more. Odevixibat is sold as Bylvay. In the United States it is indicated for cholestatic itch in Alagille syndrome from twelve months of age, at one hundred and twenty micrograms per kilogram once daily in the morning with a meal. It comes as oral pellets for children under nineteen and a half kilograms and as capsules above that.

Both labels share the same contraindication: prior or active hepatic decompensation, meaning variceal bleeding, ascites or hepatic encephalopathy. Both labels also say to separate the drug from bile acid binding resins such as cholestyramine by at least four hours, because the resin can bind the drug in the gut. There has never been a head-to-head trial, and the labelled doses are not equivalent across the two molecules. So the choice between them is made by the child's age, the formulation, access and tolerance, not by any proven difference in efficacy.

The first trial missed its target

The story begins with a trial that failed. The ITCH trial, run by the North American Childhood Liver Disease Research Network and published by Benjamin Shneider and colleagues in Hepatology Communications in 2018, randomised thirty-seven children with Alagille syndrome to placebo or to maralixibat at seventy, one hundred and forty or two hundred and eighty micrograms per kilogram per day for thirteen weeks. The prespecified primary analysis, which combined the two higher doses, showed a fall in caregiver-rated itch only about half a point greater than placebo, and this was not statistically significant. Oddly, the seventy and one hundred and forty microgram doses each showed a difference of about nine tenths of a point, which was nominally significant, while the two hundred and eighty microgram dose showed almost nothing. All doses pooled gave about six tenths of a point, just short of significance. A fall of one point or more was seen in sixty-five percent of treated children by caregiver report against twenty-five percent on placebo, and by clinician score in seventy-six percent against twenty-five percent. Serum bile acids and liver tests did not change significantly.

The honest reading is that ITCH gave a signal but not proof. The doses were low and thirteen weeks was short. The next trial raised the dose to three hundred and eighty micrograms per kilogram and changed the design completely.

ICONIC: give the drug to everyone, then take it away

The ICONIC trial was published by Emmanuel Gonzales and colleagues in The Lancet in 2021. Thirty-one children aged one to eighteen years, with a mean age of about five and a half, took open-label maralixibat, escalated to three hundred and eighty micrograms per kilogram once daily, for eighteen weeks. Then twenty-nine of them were randomised, in double-blind fashion, to stay on the drug or to switch to placebo for four weeks. After that everyone went back on open-label drug, and twenty-eight were analysed at week forty-eight. In a long-term extension, the dose could be raised to three hundred and eighty micrograms per kilogram twice daily, and fifteen children were followed to week two hundred and four, almost four years.

Why use this design? In a rare disease with severe itch, few families will accept months of placebo. A withdrawal design asks a narrower question: does the benefit depend on the drug? If the itch and the bile acids come back when the drug is stopped, the drug was doing the work.

They did come back. In the children switched to placebo, serum bile acids rose by about ninety-four micromoles per litre, and the difference between the drug and placebo groups during that window was about one hundred and seventeen micromoles per litre in favour of the drug, which met the primary endpoint. The itch score rose by about one point seven on the zero to four scale in the children switched to placebo. Over the whole period from baseline to week forty-eight, bile acids fell by about ninety-six micromoles per litre and the itch score fell by about one point six. The benefit held in the children followed to almost four years. Side effects were mostly gastrointestinal, mild to moderate and self-limiting.

So ICONIC proves that the effect depends on the drug. It does not tell you the size of a fresh effect against placebo, and the average gain, about one and a half points on a four-point scale, is a real improvement but not a cure.

ASSERT: the first conventional placebo-controlled win

The ASSERT trial, published by Nadia Ovchinsky and colleagues in The Lancet Gastroenterology and Hepatology in 2024, did what ICONIC did not. Fifty-two children with Alagille syndrome, with a median age of about five and a half years, from twenty-one centres in ten countries, were randomised two to one to odevixibat at one hundred and twenty micrograms per kilogram per day or to placebo for twenty-four weeks. Thirty-five received the drug and seventeen received placebo. The primary endpoint was the change in a caregiver-rated scratching score, also on a zero to four scale.

The scratching score fell by one point seven with odevixibat and by zero point eight with placebo. The difference, about nine tenths of a point, was statistically significant, with a p value of about zero point zero zero two. Serum bile acids fell from week four and stayed down to week twenty-four, significantly more than on placebo. Sleep improved too, with fewer nights on which the child needed help to fall asleep or slept with a caregiver. Fifty of the fifty-two children entered the open-label extension, and the gains in itch and bile acids held to at least week forty-eight.

Diarrhoea occurred in twenty-nine percent of children on odevixibat and six percent on placebo, and abdominal pain in fourteen percent against six percent. The diarrhoea was mild and did not lead anyone to stop the drug.

Notice what the placebo arm teaches. Children on placebo improved by almost a point, because of time, attention and regression of severe scores toward the average. The drug adds about one point on top of that. That is worth remembering when you judge the response in your own patient.

Does the effect last?

An integrated analysis of the maralixibat trials in Alagille syndrome, presented by Binita Kamath and colleagues at the 2024 congress of the European Association for the Study of the Liver, followed eighty-six children for up to seven years. At baseline the median observer itch score was two point seven. The median fall from baseline was one point six at year one and stayed between about one and two points through year seven, when it was two point one. Serum bile acids started at a median of about one hundred and ninety-four micromoles per litre and were lower by fifty-three at year one, by seventy-seven at year five and by one hundred and twenty-one at year seven. Height also improved slowly. The median height z-score, which measures height in standard deviations from the average for age, started at minus one point six and was higher by about zero point one at year one, zero point four at year five and zero point eight at year seven. Of thirty-eight responders followed beyond three years, thirty-six, or ninety-five percent, kept an improvement of one point or more.

There is a catch in reading these numbers. The number of children still on treatment fell from seventy-six at year one to twenty-three at year seven. Children who were transplanted or stopped the drug leave the denominator, so the later years describe survivors who were doing well. Even so, the message is that responders tend to stay responders, and growth catches up slowly.

Event-free survival and how far to trust it

The most important question for the family is whether the drug changes the fate of the liver. Two papers in Hepatology address this.

The first, by Ronald Sokol and colleagues in 2023, looked at seventy-six children treated with maralixibat for a median of four point seven years. Sixteen of them, or twenty-one percent, had a liver-related event: ten transplants, three decompensations, two deaths and one surgical biliary diversion. Response at week forty-eight predicted six-year event-free survival. Children whose itch fell by more than one point had six-year event-free survival of eighty-eight percent, against fifty-seven percent in those whose itch did not. Children whose bilirubin at week forty-eight was below six and a half milligrams per decilitre had ninety percent event-free survival, against forty-three percent above that level. Children whose bile acids were below two hundred micromoles per litre had eighty-five percent, against forty-nine percent. Itch improved in sixty-one percent of children, bilirubin in sixty-eight percent and bile acids in seventy-four percent. Starting before thirty-six months of age was linked to worse outcomes, probably because the sickest infants were the ones enrolled early.

The second paper, by Bettina Hansen and colleagues in 2024, compared eighty-four maralixibat-treated trial participants with children from the Global Alagille Alliance registry, known as GALA. Of more than fourteen hundred GALA children, four hundred and sixty-nine met the key trial eligibility criteria. The authors used a statistical method to choose, for each registry child, the point in time that best matched the disease severity of the trial children at the start of treatment. Age, bilirubin, gamma-glutamyl transferase and alanine aminotransferase were then balanced between the groups. An event was defined as variceal bleeding or ascites needing treatment, surgical biliary diversion, liver transplant, or death. Over up to six years, the hazard ratio for an event was about zero point three, meaning roughly a seventy percent lower rate of events in the maralixibat group, with a confidence interval from about zero point one nine to zero point four nine. Transplant-free survival gave a similar adjusted hazard ratio of about zero point three three.

How far should you lean on this? There are four reasons for caution. First, selection: trial children were itchy, compensated, motivated and closely followed, and registry children were not chosen this way. Serum bile acids are rarely measured in routine clinics, so the groups could not be matched on them. Second, era: the GALA children were born between 1997 and 2019, and transplant thresholds and supportive care change over time. Third, the endpoint itself: a transplant done for itch is a decision. A drug that relieves the itch removes the reason for that transplant, whether or not it slows the liver disease. In GALA, sixty-nine percent of transplants were done for intractable itch. Fourth, the analysis was funded by the manufacturer. On the other side, the results are consistent: responders by itch, by bilirubin and by bile acids all did better, which fits a real biological effect.

The honest position is this. Tell families the drug clearly reduces itch and bile acids, and is associated with fewer transplants and liver events over six years. Do not promise it will save the liver. The comparison cannot separate fewer transplants for itch from slower liver disease, and no randomised trial has shown disease modification.

What the native liver does without the drug

To weigh any of this, you need the natural history. The GALA study group, with Shannon Vandriel and Binita Kamath among the leading authors, described fourteen hundred and thirty-three children from sixty-seven centres in twenty-nine countries, born between 1997 and 2019, in Hepatology in 2023. Native liver survival was about fifty-four percent at ten years and about forty percent at eighteen years. In other words, most children with Alagille syndrome lose their native liver by adulthood.

The bilirubin in infancy separates the risk. The authors used the median total bilirubin between six and twelve months of age. Compared with children whose bilirubin was below five milligrams per decilitre, those between five and ten had about four times the risk of clinically evident portal hypertension, and those at ten or above had about eight times the risk. Seventy-nine percent of children with a bilirubin below five reached adulthood with their own liver. So put the infant's bilirubin next to the decision. A high early bilirubin means more to gain, and a closer watch for the point at which transplant becomes the right answer.

Safety and monitoring

The adverse effects follow the mechanism. In the pooled maralixibat Alagille trials described in the United States label, which included eighty-six children and had no control group, diarrhoea was reported in about fifty-six percent, abdominal pain in about fifty-four percent and vomiting in about forty-one percent. Fat-soluble vitamin deficiency was reported in about twenty-six percent, a rise in liver enzymes in about nineteen percent and bone fractures in about nine percent. Long open-label exposure inflates these rates, which is why the placebo-controlled ASSERT figure, diarrhoea in twenty-nine percent against six percent, is the better estimate of the true drug effect. Dose reduction or interruption for gastrointestinal effects was needed in only about six percent.

Three safety issues deserve deliberate attention. The first is the liver. Drug-induced liver injury is a labelled risk for both drugs. Because liver tests in Alagille syndrome are always abnormal, the trap is to miss a new rise. Obtain baseline liver tests, including alanine aminotransferase, aspartate aminotransferase and bilirubin, and check them frequently for the first six to eight months. For a new rise, reduce or interrupt the dose, restart at the last tolerated dose once the tests return to baseline, and stop for good if the abnormality recurs or the liver decompensates. The second is fat-soluble vitamins. More bile acid in the stool means less vitamin A, D, E and K absorbed, and both fractures and bleeding have occurred. Check vitamins A, D and E and the international normalised ratio, or INR, at baseline and through treatment, supplement as needed, and consider stopping if the deficiency persists despite supplements. The third is specific to the maralixibat solution, which contains propylene glycol. In children under five, watch for signs of propylene glycol toxicity.

Beyond the labels, a sensible monitoring plan includes a two-week itch diary before starting and during treatment, with sleep recorded, so that a fall of one point or more can be judged. Bile acids, cholesterol, xanthomas and height are worth tracking every three to six months. At the start, check other drugs: separate any resin by four hours, and for maralixibat, which inhibits a

drug transporter called OATP2B1 in laboratory studies, review substrates such as statins. Finally, any decompensation, whether a variceal bleed, ascites or encephalopathy, ends the drug. That is a label contraindication, and it is also the point at which the transplant decision takes over.

When to start

Putting the labels, the guideline and the outcome data together gives a practical sequence, though it is not a graded algorithm. First, confirm the diagnosis, ideally with a JAG1 or NOTCH2 variant, confirm that bile reaches the gut, and confirm that there has been no decompensation. Second, score the itch with a two-week diary and record sleep, xanthomas, baseline bile acids, vitamins and liver tests. Third, start the drug when the child has cholestatic itch that matters to the family. Start early, not after years of the old ladder. Fourth, reassess the diary at about three months and the laboratory tests at six months. Keep rifampicin only if it seems to add something on top. Fifth, at week forty-eight, look at the three responses that predicted outcome: itch, bilirubin and bile acids. A child with none of these responses needs the transplant conversation.

Two limits should be stated plainly. There are no data for starting an IBAT inhibitor in a child without itch in the hope of protecting the liver. And there are no data for infants under three months of age. The three-month and week-forty-eight checkpoints are reasonable practice drawn from the timing of the trials, not tested rules.

What the guidelines say

The 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver diseases says that, when available, IBAT inhibitor treatment should be offered to patients with Alagille syndrome and cholestatic itch. This is a strong recommendation based on level two evidence, and it effectively moves these drugs ahead of the older rungs of the ladder. The same guideline says that liver transplantation remains the standard of care for progressive cholestasis in Alagille syndrome, and it does not endorse IBAT inhibitors to prolong native liver survival. It gives no specific monitoring advice for these drugs.

The United States labels agree on the core indication, cholestatic itch in Alagille syndrome, but differ in age: three months for maralixibat and twelve months for odevixibat. They are the only source of formal monitoring advice. The North American and European pediatric societies' 2017 guideline on cholestatic jaundice in infants predates these drugs and does not address them, and the American Association for the Study of Liver Diseases has no pediatric Alagille guidance. So the comparison comes out as follows. On using the drug for Alagille itch, the documents are aligned. On age, they differ. Neither prefers one drug over the other. On the place in the ladder and on use to prolong native liver survival, there are material gaps, because only Europe grades the decision and no one endorses disease modification. On monitoring, there is a gap filled only by the labels.

When the drug does not seem to work

Before calling an IBAT inhibitor a failure, check the basics. The first is adherence. The drug must be given daily, at the right time and in the right weight-based volume, and gastrointestinal effects make families skip doses. The second is resin timing. Cholestyramine binds the drug, so the two must be given at least four hours apart, or the resin stopped. The third is dose. The ICONIC extension allowed maralixibat at three hundred and eighty micrograms per kilogram twice daily, but dosing above the label is a specialist decision.

Switching is the newest question. At The Liver Meeting in 2026, Nadia Ovchinsky and colleagues described a retrospective review in which three adolescents with Alagille syndrome were moved from maralixibat to odevixibat, two because of gastrointestinal side effects and one because of poor adherence and uncontrolled itch. All three itched less after the switch, and the diarrhoea and abdominal pain that occurred on odevixibat were transient. Failure of one IBAT inhibitor may therefore not mean failure of the class. But three uncontrolled patients prove nothing. Fix adherence and resin timing first, and do not let a switch delay a needed transplant assessment. A persistent itch score of three or more, lost sleep, spreading xanthomas or poor growth belong at the transplant meeting.

Common traps

Several errors recur. The first is keeping the IBAT inhibitor as the last rung after years of poor sleep, which runs against the 2024 European guideline. The second is telling the family the drug will save the liver, when the survival data are a matched historical comparison. The third is giving cholestyramine in the same hour as the IBAT inhibitor and then calling the drug a failure. The fourth is starting without a baseline itch diary, so that no one can say at three months whether it worked. The fifth is not checking vitamins A, D, E and the INR, and then meeting a fracture or a bleed. The sixth is continuing the drug after a variceal bleed or ascites, when decompensation is a contraindication. The seventh is missing a drug-induced rise in liver tests because Alagille liver tests are always abnormal; always compare with the child's own baseline. The eighth is expecting a response when bile does not reach the gut. The ninth is reading the almost one-point fall seen on placebo in ASSERT as drug effect in your own patient. And the tenth is letting a switch between IBAT inhibitors delay a transplant assessment in a child who is not responding.

Back to the girl, and what to do differently in clinic

Return to the girl of two. She is compensated, bile reaches her gut, and she still scores close to three on rifampicin. She should start an IBAT inhibitor now, not after another year of the old ladder. At her age either drug is labelled in the United States. Before the first dose, her mother should keep a two-week itch diary that also records sleep, and the team should check liver tests, vitamins A, D and E and the INR. Any resin must be separated by four hours. The itch should be reassessed at about three months, and bilirubin and bile acids at week forty-eight. If her diarrhoea is troublesome, the dose can be reduced and then rebuilt. If her liver tests rise, the drug should be held and restarted at the last tolerated dose, and stopped if the rise recurs. If she ever bleeds from varices or develops ascites, the drug stops and the transplant decision takes over.

What should change in clinic is this. Stop treating the IBAT inhibitor as the last resort; in Alagille syndrome it is now the best-evidenced treatment for itch, and the European guideline says to offer it when available. Measure the itch before and during treatment, because a fall of one point on a four-point scale is the effect to look for, and placebo alone gives almost a point. Watch the liver and the vitamins on purpose. Use week forty-eight as a checkpoint, because response in itch, bilirubin and bile acids predicts six-year outcome. And talk honestly with families: the drug reliably eases itch and may help the liver, but that second promise has not yet been proven in a randomised trial.

End of chapter.